

1. Administrative & Study Identification

Full Study Title A Phase II, Randomized, Double-blind, Placebo-controlled, Multicenter Study Evaluating Efficacy, Safety, and Pharmacokinetics of Crovalimab in Patients With Antiphospholipid Syndrome (APS) **Short Title/Acronym** A Study to Evaluate Crovalimab in People With Antiphospholipid Syndrome (APS) **Protocol Number** [Internal Sponsor ID] **NCT Number** NCT07172022 **Posting ID** 974576296684220866 **Sponsor Name** Hoffmann-La Roche (Roche) **Investigational Product** Crovalimab (Complement C5 inhibitor) **Phase of Development** Phase II **Medical Monitor** Hoffmann-La Roche **Clinical Trial Manager** **Trial Status** Not Yet Recruiting

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the efficacy of crovalimab compared with placebo as an add-on therapy to vitamin K antagonist (VKA) by measuring the time from randomization to the first occurrence of objectively confirmed arterial thrombosis, venous thromboembolism, or cardiovascular death. **Secondary Objectives:** To assess specific types of thrombotic events (arterial vs. venous), monitor overall safety through the percentage of participants with adverse events graded by CTCAE v5.0, evaluate treatment discontinuation, and measure blood concentrations of crovalimab over time (pharmacokinetics).

2.2 Background & Rationale Antiphospholipid syndrome (APS) is a challenging autoimmune condition where the immune system mistakenly attacks normal blood proteins, significantly increasing the risk of life-threatening blood clots in arteries and veins. Patients often experience recurrent thrombotic events despite standard anticoagulant treatments, creating an urgent need for more effective therapies. Crovalimab targets the complement system, representing a promising approach to address the underlying immune dysfunction rather than solely managing the clotting risk, with the goal of improving the quality of life for APS patients.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (1:1 allocation to crovalimab vs. placebo)

3.2 Planned Enrollment & Duration **Target N:** 300 adult participants **Number of Sites:** Multicenter **Estimated Study Start:**

Not Yet Recruiting **Estimated Primary Completion:** ~4.5 years of follow-up per participant

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years and ≤ 70 years, and body weight ≥ 40 kilograms (kg), at the time of signing Informed Consent Form. 2. Vaccination against *N. meningitidis*, *H. influenzae* type B, and *S. pneumoniae*. 3. Participants classified with APS based on the 2023 American College of Rheumatology/European Alliance of Associations for Rheumatology (ACR/EULAR) criteria, positive for at least two of the following: lupus anticoagulant (LAC) test, anticardiolipin antibodies (aCL), anti- β_2 -glycoprotein 1 antibodies (a β_2 GP1). 4. History of at least two prior arterial and/or venous thrombotic events. 5. Participants receiving corticosteroids, antimalarial treatment, non-biologic disease-modifying rheumatic drugs, statins, and low dose aspirin must be on a stable dose prior to the first dose of study treatment. 6. Willingness and ability to comply with a VKA regimen titrated to a therapeutic target international normalized ratio (INR). 7. Agreement to adhere to the contraception requirements.

4.2 Exclusion Criteria 1. Pregnant or breastfeeding, or intention of becoming pregnant during the study or within the time frame in which contraception is required. 2. Treatment with investigational therapy, complement inhibitor, and/or other immune-suppressive biologic therapy within 5 half-lives of that agent prior to screening visit, or plans to participate in another investigational trial. 3. Presence of another systemic autoimmune disease that is unstable and requires additional treatment, and constitutes the principal illness and may impact evaluation of the concurrent APS. 4. Inadequate renal and hepatic function. 5. Uncontrolled hyperlipidemia and/or hypertension, known diabetes mellitus, and/or serious infection requiring hospitalization or antibiotics prior to Week 1 Day 1. 6. History or condition associated with increased bleeding risk.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive 3 loading doses of crovalimab (680 mg or 1020 mg, based on body weight) or matching placebo at Weeks 1, 2, and 3. This is followed by maintenance doses administered every 4 weeks (Q4W) starting from Week 5. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Continued until the occurrence of arterial thrombosis, venous thromboembolism, or cardiovascular death, or up to the ~4.5-year study end.

5.2 Concomitant & Prohibited Therapy Permitted: Standard vitamin K antagonist (VKA) therapy with a target international normalized

ratio (INR) based on the investigator's discretion and local guidance. Stable doses of corticosteroids, antimalarials, non-biologic DMARDs, statins, and low-dose aspirin are permitted. **Prohibited:** Other complement pathway inhibitors, unapproved investigational agents, or immune-suppressive biologic therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, APS Marker Confirmation, Vaccination Status, Baseline INR
Loading Phase	Weeks 1, 2, 3	Randomization, Loading SC Injections, Safety and VKA Monitoring
Maintenance	Week 5 & Q4W	Q4W SC Injections, Efficacy/Safety Assessment, Pharmacokinetics
End of Study	~4.5 Years	Final Event Adjudication, Exit Interview, Safety Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Time from randomization to the first occurrence of objectively confirmed arterial thrombosis, venous thromboembolism, or cardiovascular death. **Measurement Method:** Clinical event adjudication combined with objective imaging/radiological confirmation of thrombosis.

7.2 Secondary Endpoints 1. Time from randomization to the first occurrence of objectively confirmed venous thromboembolism. 2. Time from randomization to the first occurrence of objectively confirmed arterial thrombosis. 3. Percentage of participants with Adverse Events (AEs) with severity determined according to the Common Terminology Criteria for Adverse Events Version 5.0 (CTCAE v5.0). 4. Pharmacokinetics (PK) profile of crovalimab (blood concentrations over time).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of adverse events (graded per CTCAE v5.0), targeted tracking of injection site reactions, hypersensitivity responses, and serious infections. **Serious Adverse Events (SAEs):** Reported within standard 24-hour timelines. **Data Monitoring Committee (DMC):** Independent safety monitoring committee to oversee treatment safety and VKA bridging.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for the primary efficacy time-to-event endpoint; Safety Set for all participants

receiving at least one dose. **Sample Size Justification:** 300 participants powered to detect a clinically meaningful delay in time-to-first thrombotic/cardiovascular event over 4.5 years. **Handling of Missing Data:** Time-to-event analysis utilizing Kaplan-Meier estimates and standard censoring techniques for patients lost to follow-up or withdrawn early.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was designed to be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol adherence, correct SC administration, and strict VKA/INR tracking. **Audit Trail:** Secure EDC system with eCRF locking and validated data clarification forms.

11. Study Identifiers & Governance

Primary ID: 974576296684220866 (Posting ID) **Registry Number:** NCT07172022 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** A Study to Evaluate Crovalimab in People With Antiphospholipid Syndrome (APS) **Official Regulatory Title:** A Phase II, Randomized, Double-blind, Placebo-controlled, Multicenter Study Evaluating Efficacy, Safety, and Pharmacokinetics of Crovalimab in Patients With Antiphospholipid Syndrome (APS)

12. Clinical Framework (APS Specific)

Primary Condition: Antiphospholipid Syndrome (APS) **Diagnosis Threshold:** Positive laboratory markers (LAC, aCL, a β 2GP1) via 2023 ACR/EULAR criteria + ≥ 2 prior arterial and/or venous thrombotic events **Medical Methodology:** Add-on targeted therapy (Complement C5 inhibition) combined with standard anticoagulation **Optimization Target:** Prevention of breakthrough thrombotic events and cardiovascular death

13. Study Procedures & Methodology

Management Pathway: Standard VKA therapy plus investigational crovalimab SC injections **Education Protocol (HCP):** Training on proper SC injection procedures, hypersensitivity management, vaccination requirements, and INR monitoring adjustments. **Education Protocol (Patient):** Adherence to scheduled clinic visits, contraception protocols, and continuous compliance with concomitant VKA therapy. **Quality Audit Mechanism:** Source data

verification for objectively confirmed thrombotic events and centralized adjudication.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 4.5 years **Onsite Visit Cadence:** Weekly for the first 3 weeks, then every 4 weeks (Q4W) **Remote Monitoring Frequency:** Between dosing visits as needed for VKA INR tracking **Checklist Review Interval:** Routine monitoring of bleeding/clotting signs at each Q4W visit

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					